

Tools, Memory & The MCP Standard

The Agent's Body

The 'Frozen' Intelligence Problem

- LLMs are static snapshots of the past
 - They cannot take real-world actions (send emails or trade stocks)
 - The Bridge: Tools
 - Act as actuators for the model
 - Essentially function calls that allow the model to impact the real world



How Agents Talk to Tools

- Tools are python scripts
- LLM is just a text generator. No hands. Cannot “press run” on a python script
- The Magic really is in the LLM understanding the list of tools and correctly formatting the tool call, that is, the function call.

Agent

Agent Prompt

You are a helpful assistant. You have access to a tool called `get_weather` which takes one argument: `city` (string). If you need to use it, output JSON in this format: `{"action": "tool_name", "action_input": "arguments"}`.

The Integration Nightmare

- Tools are often living on different servers (Google APIs, Slack APIs etc)
- An Agent might need to call these tools to get a job done. It might be best to allow the agent to learn the list of tools dynamically.
- Model Context Protocol (MCP)
 - Open standard for agent-tool interaction
 - Host, Client, Server architecture
 - Decouples reasoning from execution

MCP Host with an Agent

Initialization

Agent Prompt

You are a helpful assistant. You
have access to these tools:

MCP Server 

Stateless vs Stateful Agents

- LLMs are stateless (they do not have long term memory)
- Agents not having memory can become frustrating!
- We can “provide” this memory by simple “text injection”

Agents with Memory

- Adding memory to agents can help
 - Personalization
 - Context Continuity
 - Efficiency
 - Long-term reasoning
- Short term vs. Long term

Who is the secret identity of Batman?

Bruce Wayne.

Who plays him in The Dark Knight?

Christian Bale.

Who plays him in The Dark Knight?

Which character?

Agent with memory

Agent Prompt

You are a helpful assistant. You have access to a tool called save_memory. Instruction: If the user mentions a personal preferences, you MUST call save_memory to store it for the future. Do not save casual conversation, only permanent facts.

<memory.txt>